

Assignment 1Simple Linear RegressionProblem Statement

A mechanical engineer is analyzing the relationship between the applied load on a spring and the extension produced. Using the dataset given below, build a Simple linear Regression model to study the relationship between load and extension.

Dataset

Load (N)	Extension (mm)
5	1.2
10	2.5
15	3.7
20	4.9
25	6.1

Linear Regression

$$y = mx + c$$

$$m (\text{slope}) = \frac{n \sum xy - \sum x \sum y}{n \sum (x^2) - (\sum x)^2}$$

where $m \rightarrow$ slope

$n \rightarrow$ number of data

$x \rightarrow$ Load in newton (N)

$y \rightarrow$ Extension in millimeter (mm)

Here $n = 5$

x	y	xy	x^2
5	1.2	6	25
10	2.5	25	100
15	3.7	55.5	225
20	4.9	98	400
25	6.1	152.5	625

$$\sum x = 75$$

$$\sum y = 18.4$$

$$\sum xy = 337$$

$$\sum x^2 = 1375$$

$$(\sum x)^2 = 5625$$

$$m = \frac{n \sum xy - \sum x \sum y}{n \sum (x^2) - (\sum x)^2}$$

$$m = \frac{(5 \times 337) - (75 \times 18.4)}{(5 \times 1375) - 5625}$$

$$m = \frac{1685 - 1380}{6875 - 5625}$$

$$m = 0.244$$

$$c = \frac{\sum y - m \sum x}{n}$$

$$c = \frac{18.4 - (0.244 \times 75)}{5} = 0.02$$

$$c = 0.02$$

$$\therefore y = mx + c$$

$$\therefore y = 0.244x + 0.02$$

$$y = 0.244x + 0.02 \rightarrow \text{ML model}$$

Extension when the applied load is 55 N

$$x = 55 \text{ in N}$$

y = required extension in mm

$$y = 0.244 \times 55 + 0.02$$

$$y = 13.4 \text{ mm}$$

$$\therefore \text{Required extension} = 13.4 \text{ mm}$$

$$\text{Extension (y)} = 13.4 \text{ mm}$$

File Edit Selection View Go Run Terminal Help

Untitled (Workspace)

EXPLORER

UNTITLED (WORKSPACE)

coding

New folder

data.csv

LoadExtension_ALPHIN.py

LoadExtension.py

ml_program.py

coding > LoadExtension_ALPHIN.py > ...

```
1
2 import pandas as pd
3 import matplotlib.pyplot as plt
4 from sklearn.linear_model import LinearRegression
5 data = pd.read_csv("data.csv")
6 x = data[["Load (N)"]]
7 y = data[["Extension (mm)"]]
8
9 plt.scatter(x,y)
10 plt.show()
11 model = LinearRegression()
12 model.fit(x,y)
13 print("Coeff: ",model.coef_)
14 print("Intercept: ",model.intercept_)
15
16 load_1 = 55
17 extension_1 = model.predict([[load_1]])
18 print("The extension for load 55 is: ", extension_1)
19
20 load = int(input("Enter the load: "))
21 extension = model.predict([[load]])
22 print("The extension is: ", extension)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

The extension is: [[19.052]]
PS D:\coding> & C:\Users\alphi\AppData\Local\Programs\Python\Python314\python.exe d:/coding/LoadExtension_ALPHIN.py
Coeff: [[0.244]]
Intercept: [0.02]
C:\Users\alphi\AppData\Local\Programs\Python\Python314\Lib\site-packages\sklearn\utils\validation.py:2691: UserWarning: X does not have valid feature names, but LinearRegression was fitted with feature names
warnings.warn(
The extension for load 55 is: [[13.44]]
Enter the load: 75
C:\Users\alphi\AppData\Local\Programs\Python\Python314\Lib\site-packages\sklearn\utils\validation.py:2691: UserWarning: X does not have valid feature names, but LinearRegression was fitted with feature names
warnings.warn(
The extension is: [[18.32]]
PS D:\coding> |

power...
power...
powershell
Python

Tip: Try the Plan agent to research and plan before implementing changes.

+ LoadExtension_ALPHIN.py

Describe what to build

+ Auto

Local Default Ap...

Ln 14, Col 38 Spaces: 4 UTF-8 CRLF Python Python 3.14 (64-bit)

22:23 19-05-2026

